XML cleanup and its preservation guards

Abbreviated source examples, negative cases and native rendering evidence

This atlas groups the reviewed behavior specification, implementation, commit history and focused regressions into 50 visual examples. Production Default rules and explicitly labeled diagnostic fixtures have separate scopes. Experimental Maximum and Too Aggressive settings have different preservation policies.

Read each source/result pair at its native one-column width. Red backgrounds and minus signs identify removed lines on the right. Green backgrounds and plus signs identify added lines on the left. An equals sign identifies a protected structure. Folded color bands connect edited ranges to thin markers at their exact gaps. These cues supplement syntax color.

The XML is illustrative. Namespace wrappers, irrelevant attributes, cached payloads and repeated cells are omitted. Style indices have meaning only within their own example. Comments describe omitted scope or an exclusion. A red omission label stands for deleted records. Explanatory comments are not literal output. The complete native font-control packages are retained separately.

The source index pins each case to a file and a regression-test or implementation anchor. A matching unit test checks its own fixture. It does not prove that an abbreviated picture is a complete runnable workbook or that every possible Excel presentation is preserved.

Native evidence. The reduced font control has three native Excel vector exports after CalculateFullRebuild. The Enron fill comparison is re-exported from the retained rebuild/Save controls. All other outcome descriptions are traced to their named tests or investigation records. No schematic is presented as an Excel screenshot.

Boundaries. Unsupported XML can exclude an input. Unknown inheritance or references follow the local conservative guard. Ordinary hidden tables are not covered by hidden-tail admission. Alias-name cleanup does not currently prove equality of formula bodies. Selected-region native equality is not whole-workbook certification.

Reviewed source revision 1b9ee98e3409c3584ffd12a661d0c19f8a5f0788. Core verification completed with 298 passed, zero failed and one manual timing test ignored. The source repository was read and tested for this figure task. Cleanup behavior was not edited.

The design follows Animated Vega-Lite Figure 2 and Bluefish Figure 2. This atlas uses newly drawn geometry, editable text, local diff highlighting and native control images.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

01 Content is not formatting

Source

```
<row r="1">
 <c r="A1"><v>0</v></c>
 <c r="B1"><f t="shared" si="0"/></c>
 <c r="C1">
  <v xml:space="preserve">7</v></c>
</row>
```

=

Retained by the guard

```
<row r="1">
 <c r="A1"><v>0</v></c>
 <c r="B1"><f t="shared" si="0"/></c>
 <c r="C1">
  <v xml:space="preserve">7</v></c>
</row>
```

Keep zero, shared-formula followers and attributed value elements. Empty cached display is insufficient.

Unsafe shortcut. Deleting a zero or a formula changes workbook data.

Abbreviated XML. Style IDs are local.

02 Keep one font witness

Source

```
<row r="2">
 <c r="A2">...</c>
 <c r="B2" s="1"/>
- <c r="C2" s="1"/>
</row>
```

Guarded result

```
<row r="2">
 <c r="A2">...</c>
 <c r="B2" s="1"/>
</row>
```

A2 uses Arial 10. B2 and C2 carry Arial 18. Keep one needed font representative in this row.

Unsafe shortcut. Removing both witnesses shortens this controlled Excel row from 23.25 to 12.75 points.

Abbreviated XML. Style IDs are local.

03 A witness is local to a merge

Source

```
<row r="6" ht="177"
 customHeight="1">
 <c r="A6" s="237">...</c>
 <c r="B6" s="238"/>
- <c r="C6" s="238"/>
- <c r="H6" s="238"/>
</row>
<mergeCell ref="A6:H6"/>
```

Guarded result

```
<row r="6" ht="177"
 customHeight="1">
 <c r="A6" s="237">...</c>
 <c r="B6" s="238"/>
</row>
<mergeCell ref="A6:H6"/>
```

Retain one representative per encountered font in each merged area and row, even at fixed height.

Unsafe shortcut. Deleting B6:H6 changed justified paragraph spacing. A witness in another merge is insufficient.

Abbreviated XML. Style IDs are local.

04 No fill can be a visible override

Source

```
<row r="6" s="2"
 customFormat="1" ht="18"
 customHeight="1">
 <c r="K6"/>
 <c r="L6"/>
</row>
```

=

Retained by the guard

```
<row r="6" s="2"
 customFormat="1" ht="18"
 customHeight="1">
 <c r="K6"/>
 <c r="L6"/>
</row>
```

Row style 2 has a blue fill. The explicit styleless cells clear it. Keep every local mask, not one font representative.

Unsafe shortcut. Deleting K6 exposes blue at K6, even if L6 remains.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

05 Remove an override with nothing to clear

Source

```
- <row r="2" ht="12.75"
-  customHeight="1">
-  <c r="A2" s="5"/>
-  <c r="B2" s="5"/>
- </row>
```

Guarded result

```
  <!-- Content-free tail removed -->
```

Style 5 applies fillId=0. No visible inherited fill or border exists and no other retention rule applies.

Unsafe shortcut. The same attributes must survive when a white or colored parent fill is present.

Abbreviated XML. Style IDs are local.

06 An implicit row style still overrides

Source

```
 <col min="6" max="6" style="17"/>
 <row r="187" customFormat="1">
  <c r="F187" s="17"/>
 </row>
```

=

Retained by the guard

```
 <col min="6" max="6" style="17"/>
 <row r="187" customFormat="1">
  <c r="F187" s="17"/>
 </row>
```

customFormat=1 without s applies row XF 0. A gray cell may restore a column fill that the row format cancels.

Unsafe shortcut. Removing F187 because it matches the column exposes the row no-fill format.

Abbreviated XML. Style IDs are local.

07 Font witnesses cannot replace local masks

Source

```
<col min="6" max="8"
 hidden="1" style="13"/>
<row r="14">
 <c r="A14" s="26">...</c>
 <c r="F14" s="26"/>
 <c r="G14" s="26"/>
</row>
```

=

Retained by the guard

```
<col min="6" max="8"
 hidden="1" style="13"/>
<row r="14">
 <c r="A14" s="26">...</c>
 <c r="F14" s="26"/>
 <c r="G14" s="26"/>
</row>
```

Hidden columns can affect automatic height. Preserve each mask that would expose a font absent from the original row.

Unsafe shortcut. A matching visible witness cannot stop a larger inherited font at F14 from affecting the row.

Abbreviated XML. Style IDs are local.

08 Move repeated formatting into the row

Source

```
 <row r="1" ht="9"
  customHeight="1"
 >
  <c r="A1" s="3"><v>7</v></c>
-  <c r="B1" s="2"/>
-  <!-- repeated through XFD1 -->
-  <c r="XFD1" s="2"/>
 </row>
```

Guarded result

```
 <row r="1" ht="9"
  customHeight="1"
+  s="2" customFormat="1"
 >
  <c r="A1" s="3"><v>7</v></c>
 </row>
```

Reuse an existing XF. Require a visible fixed-height, unprotected row and proved inheritance. Dense route needs at least 1,024 occurrences.

Unsafe shortcut. A newly inherited fill, border, protection or font at a gap can change the workbook.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

09 Restore necessary exceptions at gaps

Source

```
 <row r="1" ht="9"
  customHeight="1"
 >
  <c r="A1" s="3"><v>7</v></c>
-  <!-- repeated s=2 cells -->
  <!-- E1 was absent -->
 </row>
```

Guarded result

```
 <row r="1" ht="9"
  customHeight="1"
+  s="2" customFormat="1"
 >
  <c r="A1" s="3"><v>7</v></c>
+  <c r="E1" s="3"/>
 </row>
```

The bounded dense-gap route admits at most 64 missing columns and restores original formats only when the inheritance proof succeeds.

Unsafe shortcut. Promoting the row without an E1 exception exposes a new inherited font or fill.

Abbreviated XML. Style IDs are local.

10 A cached height is not a fixed height

Source

```
 <row r="105">
  <c r="A105" s="2">...</c>
  <!-- many s=2 blanks -->
 </row>
```

=

Retained by the guard

```
 <row r="105">
  <c r="A105" s="2">...</c>
  <!-- many s=2 blanks -->
 </row>
```

Do not promote an automatic row. Both ht and customHeight=1 are required, even when the proposed font is already present.

Unsafe shortcut. A historical promotion shortened Destatis row 105 from 12.75 to 9 points and moved the footer.

Abbreviated XML. Style IDs are local.

11 A centering span needs every blank

Source

```
 <row r="2">
  <c r="D2" s="9">...</c>
  <c r="E2" s="9"/>
  <c r="F2" s="9"/>
  <c r="G2" s="9"/>
 </row>
```

=

Retained by the guard

```
 <row r="2">
  <c r="D2" s="9">...</c>
  <c r="E2" s="9"/>
  <c r="F2" s="9"/>
  <c r="G2" s="9"/>
 </row>
```

Style 9 has horizontal=centerContinuous. A value or formula seeds a contiguous same-alignment span.

Unsafe shortcut. Keeping only G2 does not restore centering after F2 is deleted. A gap ends the span.

Abbreviated XML. Style IDs are local.

12 Protection does not erase a font effect

Source

```
 <row r="2">
  <c r="A2">...</c>
  <c r="B2" s="7"/>
-  <c r="C2" s="7"/>
 </row>
```

Guarded result

```
 <row r="2">
  <c r="A2">...</c>
  <c r="B2" s="7"/>
 </row>
```

Style 7 contains a font and protection metadata. Outside the protection-retention area it may still supply required automatic height.

Unsafe shortcut. Treating protection-only eligibility as permission to remove its font can shift later content.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

13 Prune only a genuinely redundant fill

Source

```
  <col min="1" max="16384" style="1"/>
  <row r="486"><c r="E486">...</c></row>
- <row r="487" s="2" customFormat="1"/>
  <!-- same tail through row 49633 -->
```

Guarded result

```
  <col min="1" max="16384" style="1"/>
  <row r="486"><c r="E486">...</c></row>
```

Column style 1 covers A:XFD with white. Tail row style 2 repeats that fill and has no remaining anchor. Check exact column overrides and gaps.

Unsafe shortcut. A blue tail on the same white columns remains visible. White fills also hide gridlines.

Abbreviated XML. Style IDs are local.

14 A font alone need not anchor a tail

Source

```
- <row r="1424" s="6"
-  customFormat="1">
-  <c r="A1424"/>
-  <c r="B1424" s="5"/>
- </row>
```

Guarded result

```
  <!-- Empty sheetData -->
```

There are no values, visible fills, borders, merges, drawings or later anchors. Font-only representatives do not retain an otherwise content-free sheet.

Unsafe shortcut. The same row before later visible content can be needed for layout and must be reconsidered in context.

Abbreviated XML. Style IDs are local.

15 A row header can clear column borders

Source

```
  <col min="6" max="6" style="4"/>
  <row r="2" s="7" customFormat="1"/>
```

=

Retained by the guard

```
  <col min="6" max="6" style="4"/>
  <row r="2" s="7" customFormat="1"/>
```

Column style 4 has a border. Row style 7 has explicit borderId=0. Keep the masking row header, including when applyBorder is absent.

Unsafe shortcut. Deleting the empty header exposes vertical column borders across that row.

Abbreviated XML. Style IDs are local.

16 Empty row heights position visible borders

Source

```
  <col min="6" max="6" style="4"/>
  <row r="20" ht="24"/>
  <row r="21" ht="12.75"/>
```

=

Retained by the guard

```
  <col min="6" max="6" style="4"/>
  <row r="20" ht="24"/>
  <row r="21" ht="12.75"/>
```

Visible column borders make stored heights of nonhidden tail rows observable. Preserve them even without customHeight.

Unsafe shortcut. Removing an empty height record changes the extent and segmentation of the visible borders.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

17 Conditional ranges protect row layout

Source

```
  <row r="2" ht="17.25"
-  customHeight="1">
-  <c r="H2" s="6"/></row>
- <row r="60" ht="20" customHeight="1"/>
  <conditionalFormatting sqref="A5:B44">
   ...
  </conditionalFormatting>
```

Guarded result

```
  <row r="2" ht="17.25"
+  customHeight="1"/>
  <conditionalFormatting sqref="A5:B44">
   ...
  </conditionalFormatting>
```

Keep layout through the conditional region. Remove unnecessary font-only cell payloads and unrelated later tail headers separately.

Unsafe shortcut. Deleting preceding row heights shifts the end of a conditional fill, even with no ordinary cell fill.

Abbreviated XML. Style IDs are local.

18 Objects extend the layout boundary

Source

```
<row r="1"><c r="A1"><v>7</v></c></row>
<row r="50" ht="30" customHeight="1"/>
<drawing r:id="rId1"/>
<!-- drawing anchor ends at row 60 -->
```

=

Guarded result

```
<row r="1"><c r="A1"><v>7</v></c></row>
<row r="50" ht="30" customHeight="1"/>
<drawing r:id="rId1"/>
```

Resolve supported drawing relationships and anchors before trimming row layout beyond the last data cell.

Unsafe shortcut. Deleting row 50 because A1 is the last value can move or resize the later object.

Abbreviated XML. Style IDs are local.

19 Merge tail columns with font witnesses

Source

```
  <col min="10" max="20" style="1"/>
- <col min="21" max="16384" style="2"/>
```

Guarded result

```
  <col min="10" max="20" style="1"/>
+ <col min="21" max="21" style="2"/>
+ <col min="22" max="16384" style="1"/>
```

This is a schematic equal-fill tail merge. Keep one original empty-tail column per font identity that replacement would otherwise erase.

Unsafe shortcut. Replacing every Calibri 11 column with Arial 10 shortened NLA data rows from 15.75 to 15 pt.

Abbreviated XML. Style IDs are local.

20 Hidden constants need deferred approval

Source

```
<row r="8" hidden="1">
 <!-- >=4096 contiguous integer 0s -->
 <c r="A8"><v>0</v></c>
 ...
 <c r="XFD8"><v>0</v></c>
</row>
```

Decision

```
<!-- Admit suffix only after -->
<!-- reference and layout checks. -->
```

Admit only a dense horizontal suffix of equal signed integers ending at XFD, with every source cell eligible. Any intersecting known reference retains the entire span.

Unsafe shortcut. Arbitrary hidden tables, short runs, gaps, text, formulas and vertical repeats are not admitted.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

21 Translate shared references by axis

Source

```
<c r="D166"><f t="shared" si="0"
 ref="D166:J166">D91/D64*100-100</f></c>
<c r="E166"><f t="shared" si="0"/></c>
<!-- hidden E64 = 4232.35 -->
```

=

Retained by the guard

```
<c r="D166"><f t="shared" si="0"
 ref="D166:J166">D91/D64*100-100</f></c>
<c r="E166"><f t="shared" si="0"/></c>
<!-- hidden E64 = 4232.35 -->
```

Translate relative axes over the follower range. Absolute axes remain fixed. Protect enclosing rectangles without enumerating every formula. In this example the protected input rectangles include D91:J91 and D64:J64.

Unsafe shortcut. Scanning only the anchor misses E64 and can produce DIV/0 errors after rebuilding.

Abbreviated XML. Style IDs are local.

22 A blank chart-source cell carries format

Source

```
<c r="A1" s="4"/>
<c r="A2"><v>0.25</v></c>
<!-- chart series references A1:A2 -->
<!-- axis format is source-linked -->
```

=

Guarded result

```
<c r="A1" s="4"/>
<c r="A2"><v>0.25</v></c>
```

Retain the first explicit cell of a directly resolved chart series, including its original number format. Do not retain every blank in the series.

Unsafe shortcut. Deleting A1 can change a source-linked chart axis even though A1 has no value.

Abbreviated XML. Style IDs are local.

23 Whitespace can block text overflow

Source

```
<c r="A1" t="inlineStr">
 <is><t>Long heading</t></is>
</c>
<c r="B1" t="s"><v>4</v></c>
<!-- shared string 4 is one space -->
```

=

Guarded result

```
<c r="A1" t="inlineStr">
 <is><t>Long heading</t></is>
</c>
<c r="B1" t="s"><v>4</v></c>
```

Default leaves optional whitespace-value deletion disabled. A space is a value and can stop neighboring text overflow.

Unsafe shortcut. Removing B1 can expose text in the next column or affect dependencies and row sizing.

Abbreviated XML. Style IDs are local.

24 Metadata has its own predicates

Source

```
- <definedName name="Print_Area"
-  localSheetId="0">S!$A$1</definedName>
  <definedName name="_xlnm.Print_Area"
   localSheetId="0">S!$A$1</definedName>
```

Guarded result

```
  <definedName name="_xlnm.Print_Area"
   localSheetId="0">S!$A$1</definedName>
```

Match a built-in alias and its prefixed counterpart in the same raw local scope. Keep names on different sheets distinct.

Unsafe shortcut. The current predicate does not compare formula bodies. This example must not imply general semantic equivalence.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

25 Column widths position repeated borders

Source

```
<col min="8" max="8" width="4"/>
<col min="9" max="9" width="20"/>
<row r="2" s="1" customFormat="1"/>
```

=

Retained by the guard

```
<col min="8" max="8" width="4"/>
<col min="9" max="9" width="20"/>
<row r="2" s="1" customFormat="1"/>
```

A surviving row border repeats at column boundaries. Preserve existing widths even at empty positions.

Unsafe shortcut. Merging the empty columns moves the repeated border boundaries.

Abbreviated XML. Style IDs are local.

26 Compare font records, not just IDs

Source

```
  <c r="A1" s="1"><v>7</v></c>
  <c r="B1" s="1"/>
- <c r="C1" s="1"/>
  <!-- column uses another font ID -->
```

Guarded result

```
  <c r="A1" s="1"><v>7</v></c>
  <c r="B1" s="1"/>
```

Byte-identical font records share an internal identity. Retain the required representative without preserving duplicates merely for different font IDs.

Unsafe shortcut. Different font records retain their independent masks. Other XF components still matter.

Abbreviated XML. Style IDs are local.

27 Unsupported input is an exclusion

Source

```
<worksheet xmlns:s="...">
 <s:sheetData>...</s:sheetData>
</worksheet>
```

Decision

```
<!-- Excluded with part and reason. -->
<!-- No cleaned workbook published. -->
```

The fast parser targets unprefixed Transitional SpreadsheetML. Check bounded root headers, then namespace declarations during the normal stream.

Unsafe shortcut. An unsupported or malformed representation cannot be treated as empty and rewritten.

Abbreviated XML. Style IDs are local.

28 Prune stale chain entries by sheet

Source

```
  <calcChain>
   <c r="A1" i="1"/>
-  <c r="C1" i="1"/>
   <c r="C1" i="2"/>
  </calcChain>
```

Guarded result

```
  <calcChain>
   <c r="A1" i="1"/>
   <c r="C1" i="2"/>
  </calcChain>
```

Only sheet 1 was processed and its C1 record was removed. Resolve chain sheet IDs through workbook relationships.

Unsafe shortcut. Do not delete a same-coordinate entry on an unprocessed sheet or when its mapping is unknown.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

29 Trim spans without moving coordinates

Source

```
- <row r="3" spans="1:6">
    <c r="A3" s="8">...</c>
    <c r="B3" s="13"/>
-   <c r="C3" s="13"/>
-   <c r="F3" s="13"/>
  </row>
```

Guarded result

```
+ <row r="3" spans="1:2">
    <c r="A3" s="8">...</c>
    <c r="B3" s="13"/>
  </row>
```

Keep the value and required style representative. Reduce spans to this row's last retained column. Remaining cell coordinates stay unchanged.

Unsafe shortcut. Stale state from an earlier cleanup can leave an obsolete Z-column span. Deletion does not shift retained cells left.

Abbreviated XML. Style IDs are local.

30 Remove cells from an empty fixed row

Source

```
  <row r="15" s="429"
   customFormat="1" ht="27"
-  customHeight="1">
-  <c r="A15" s="427"/>
-  <c r="B15" s="429"/>
- </row>
```

Guarded result

```
  <row r="15" s="429"
   customFormat="1" ht="27"
+  customHeight="1"/>
```

The cells have only unneeded font formatting. The fixed row header already preserves its height and row format.

Unsafe shortcut. Fixed height does not authorize removal of a fill mask, merged-text witness or baseline-sensitive font.

Abbreviated XML. Style IDs are local.

31 Fixed height still needs a font witness

Source

```
  <row r="134" ht="9"
   customHeight="1">
   <c r="A134" s="7"><v>7896</v></c>
   <c r="L134" s="9"/>
-  <c r="M134" s="9"/>
  </row>
```

Guarded result

```
  <row r="134" ht="9"
   customHeight="1">
   <c r="A134" s="7"><v>7896</v></c>
   <c r="L134" s="9"/>
  </row>
```

Keep one explicit default-font witness in this text row, even though the height is fixed. Style 9 also has an explicit no-fill component.

Unsafe shortcut. Removing both blanks changes the text baseline. A fixed height alone does not prove identical text positioning.

Abbreviated XML. Style IDs are local.

32 Sparse promotion reuses a font-only XF

Source

```
  <row r="1" ht="9"
   customHeight="1"
  >
    <c r="A1" s="2">...</c>
-   <c r="B1" s="2"/>
-   <c r="C1" s="2"/>
-   <c r="D1" s="2"/>
  </row>
```

Guarded result

```
  <row r="1" ht="9"
   customHeight="1"
+  s="1" customFormat="1"
  >
    <c r="A1" s="2">...</c>
  </row>
```

All absent positions already inherit a compatible font. Promote to existing font-only XF 1 and retain A1's richer XF 2.

Unsafe shortcut. Copying XF 2 wholesale would leak its alignment or other components into the row. A different inherited gap font blocks sparse promotion.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

33 Uniform fill promotion keeps merge cells

Source

```
 <row r="1" ht="20.25"
  customHeight="1"
 >
  <c r="A1" s="2">...</c>
  <c r="C1" s="1"/>
  <c r="D1" s="1"/>
-  <!-- other cells also use s=1 -->
 </row>
 <mergeCell ref="C1:D1"/>
```

Guarded result

```
 <row r="1" ht="20.25"
  customHeight="1"
+  s="1" customFormat="1"
 >
  <c r="A1" s="2">...</c>
  <c r="C1" s="1"/>
  <c r="D1" s="1"/>
 </row>
 <mergeCell ref="C1:D1"/>
```

The full row has the same applied fill, including its content cells. Reuse XF 1 while preserving the merge's required explicit cells.

Unsafe shortcut. A different content-cell fill, visible border, border override or protection hazard prevents this promotion.

Abbreviated XML. Style IDs are local.

34 An ineligible majority still wins

Source

```
<row r="1" ht="9" customHeight="1">
 <c r="A1" s="0"/>
 <c r="B1" s="0"/>
 <c r="C1" s="0"/>
 <c r="D1" s="1"/>
 <c r="E1" s="1"/>
</row>
```

Decision

```
<!-- Dominant XF 0 wins 3:2. -->
<!-- No row promotion. -->
```

Count all candidate styles before checking promotion eligibility. An unsuitable winning style does not make the smaller eligible group the winner.

Unsafe shortcut. Discarding XF 0 from the counts would incorrectly promote XF 1. Equal counts use the lower style ID, even when unsuitable.

Abbreviated XML. Style IDs are local.

35 Protection-only retention has a boundary

Source

```
 <row r="1"><c r="B1" s="7"/></row>
 <row r="2"><c r="C2"><v>1</v></c></row>
- <row r="3"><c r="D3" s="7"/></row>
```

Guarded result

```
 <row r="1"><c r="B1" s="7"/></row>
 <row r="2"><c r="C2"><v>1</v></c></row>
```

Default KeepInDataRange retains the protection-only B1 cell inside the later-established visible extent. D3 has no independent visual or dependency role.

Unsafe shortcut. Removing all protection-only cells loses in-range metadata. Keeping all of them can anchor a large irrelevant tail.

Abbreviated XML. Style IDs are local.

36 Coalesce a compatible equal-fill tail

Source

```
  <col min="1" max="20" style="39"/>
- <col min="21" max="239" style="39"
    width="8.77"/>
- <col min="240" max="16148" style="39"
-   width="11.55"/>
- <col min="16149" max="16384" style="39"
-   width="8.77"/>
```

Guarded result

```
  <col min="1" max="20" style="39"/>
+ <col min="21" max="16384" style="39"
    width="8.77"/>
```

Beyond the used-column boundary 20, compatible visible fills and border-free formatting permit a compact tail declaration.

Unsafe shortcut. A visible row border can expose the removed widths. Hidden columns, gaps, unknown styles and incompatible components restrict the operation.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

37 Equal fill does not cover a gap

Source

```
<col min="21" max="30" style="39"/>
<!-- column 31 has no definition -->
<col min="32" max="16384" style="39"/>
```

=

Retained by the guard

```
<col min="21" max="30" style="39"/>
<!-- column 31 has no definition -->
<col min="32" max="16384" style="39"/>
```

The two defined intervals have the same visible fill. Their uncovered column 31 prevents an equal-fill merge across that gap.

Unsafe shortcut. A single 21:XFD definition would paint the previously uncovered column. Matching endpoints are insufficient.

Abbreviated XML. Style IDs are local.

38 Hidden values can affect visible height

Source

```
<col min="1" max="1" hidden="1"/>
<row r="3">
 <c r="A3" t="inlineStr">
  <is><t>Heading
second line</t></is>
 </c>
</row>
```

=

Retained by the guard

```
<col min="1" max="1" hidden="1"/>
<row r="3">
 <c r="A3" t="inlineStr">
  <is><t>Heading
second line</t></is>
 </c>
</row>
```

A hidden column is not a sufficient deletion guard in a visible automatic-height row. Keep the content that can influence that height.

Unsafe shortcut. Reference checks alone do not protect this local layout effect. Hidden rows and fixed-height rows have separate admission checks.

Abbreviated XML. Style IDs are local.

39 Small hidden tables are retained

Source

```
<row r="11" hidden="1">
 <c r="XFC11"><v>0</v></c>
 <c r="XFD11"><v>0</v></c>
</row>
```

=

Retained by the guard

```
<row r="11" hidden="1">
 <c r="XFC11"><v>0</v></c>
 <c r="XFD11"><v>0</v></c>
</row>
```

Equal integers and an XFD endpoint do not suffice. This two-cell suffix is below the 4,096-cell admission threshold.

Unsafe shortcut. Treating any unreferenced hidden table as bloat would exceed the current Default policy. Formulas and mixed or changing values also block constant-tail admission.

Abbreviated XML. Style IDs are local.

40 Shared formulas preserve absolute axes

Source

```
<c r="B10"><f t="shared" si="7"
 ref="B10:D12">
Inputs!$E5+Inputs!F$6+Inputs!$G$7
</f></c>
<c r="D12"><f t="shared" si="7"/></c>
```

=

Retained by the guard

```
<c r="B10"><f t="shared" si="7"
 ref="B10:D12">
Inputs!$E5+Inputs!F$6+Inputs!$G$7
</f></c>
<c r="D12"><f t="shared" si="7"/></c>
```

Resolve the Inputs sheet and protect E5:E7, F6:H6 and G7 as the anchor moves over B10:D12. Fixed axes do not move.

Unsafe shortcut. Expanding both axes or binding the references to the formula's own sheet gives the wrong protected cells.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

41 Local references stay on their sheet

Source

```
<!-- formulas on worksheet Other -->
<c r="T1"><f>S41</f></c>
<c r="T2"><f>Input!S41</f></c>
```

Decision

```
<!-- Protect Other!S41. -->
<!-- Protect Input!S41 separately. -->
```

An unqualified S41 belongs to Other. Only the qualified formula protects Input!S41. Resolve local defined names in their own sheet scope too.

Unsafe shortcut. A global coordinate set would retain or delete the wrong sheet's candidate. Unknown scope needs conservative handling.

Abbreviated XML. Style IDs are local.

42 Charts and pivots have scoped consumers

Source

```
<!-- chart source -->
<c:f>Input!$B$2:$D$9</c:f>
<!-- pivot cache source -->
<worksheetSource sheet="Input"
 ref="B2:D9"/>
<!-- pivot result on Output -->
<location ref="A7:B94"/>
```

Decision

```
<!-- Protect Input!B2:D9. -->
<!-- Protect Output!A7:B94. -->
```

Extended dependency scanning resolves chart series, pivot inputs and separately mapped pivot output areas. Supported linked controls and drawing text are also consumers.

Unsafe shortcut. A chart format anchor alone does not cover these value dependencies. A pivot location without its owning sheet cannot be assigned globally.

Abbreviated XML. Style IDs are local.

43 Unresolved sources block candidate deletion

Source

```
<c:f>OFFSET(Input!A1,1,0)</c:f>
<worksheetSource name="Table1"/>
<!-- Table1 cannot be resolved here -->
```

Decision

```
<!-- Reference scan is unsafe. -->
<!-- Keep deferred candidates. -->
```

A dynamic object formula or unresolved named source prevents the required dependency proof. Retain reference-sensitive candidates.

Unsafe shortcut. A failed scan must not be interpreted as an empty reference set. Structural cleanup has its own independent guards.

Abbreviated XML. Style IDs are local.

44 Delete a value, keep its visible format

Diagnostic whitespace cleanup. Default keeps whitespace.

Source

```
 <row r="4">
  <c r="W4" s="91"><v>118</v></c>
-  <c r="X4" s="37" t="s">
-   <v>15801</v>
-  </c>
  <c r="Y4" s="91"><v>7</v></c>
 </row>
```

Guarded result

```
 <row r="4">
  <c r="W4" s="91"><v>118</v></c>
+  <c r="X4" s="37" t="s"/>
  <c r="Y4" s="91"><v>7</v></c>
 </row>
```

An already admitted, unprotected whitespace value is stripped. Re-cleaning retains the now-empty XF-37 cell because its fill is still visible.

Unsafe shortcut. Deleting the entire cell would conflate value removal with formatting removal. A font witness or dependency can prevent admission earlier.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

45 Reconsider the tail after deferred deletion

Internal deferred-candidate fixture. Admission is a separate step.

Source

```
  <row r="1"><c r="A1">...</c></row>
- <row r="41">
-  <c r="S41">...</c>
- </row>
```

Guarded result

```
  <row r="1"><c r="A1">...</c></row>
```

After an admitted candidate is removed, invalidate the old row-41 tail anchor and re-clean touched rows before trimming the newly empty tail.

Unsafe shortcut. Retaining stale anchors leaves avoidable rows. Trimming before dependency filtering could discard a still-protected value.

Abbreviated XML. Style IDs are local.

46 Unfamiliar cell metadata blocks promotion

Source

```
<row r="1" ht="9" customHeight="1">
 <c r="A1" s="2" cm="3">
  <f>1+6</f><v>7</v>
 </c>
 <c r="B1" s="2"/>
 <c r="C1" s="2"/>
</row>
```

Decision

```
<!-- No row promotion. -->
<!-- Use the normal cleanup path. -->
```

Preserve the original formula/value cell and its nonstandard attributes. The row-promotion look-ahead accepts only its supported cell representation.

Unsafe shortcut. A cheap promotion scan must not silently erase cm/vm metadata, extension attributes or an unknown explicit XF.

Abbreviated XML. Style IDs are local.

47 A visible font cannot mask another column

Source

```
<col min="1" max="2" style="2"/>
<col min="5" max="16384" style="2"/>
<row r="1" ht="12.75">
 <c r="A1" s="2"><v>1</v></c>
 <c r="C1" s="2"/>
 <c r="D1" s="2"/>
</row>
```

=

Retained by the guard

```
<col min="1" max="2" style="2"/>
<col min="5" max="16384" style="2"/>
<row r="1" ht="12.75">
 <c r="A1" s="2"><v>1</v></c>
 <c r="C1" s="2"/>
 <c r="D1" s="2"/>
</row>
```

Only C:D inherit the default font. Their explicit XF-2 cells mask it locally. Keep both when removing them would introduce a previously absent font.

Unsafe shortcut. A1 already represents font 8, but cannot suppress a default font newly exposed at C1 or D1.

Abbreviated XML. Style IDs are local.

48 Collapse only the exterior hidden-row tail

Source

```
  <row r="1"><c r="A1"><v>1</v></c></row>
- <row r="2" hidden="1"/>
- <row r="3" hidden="1"/>
  <row r="4" hidden="1"/>
```

Guarded result

```
  <row r="1"><c r="A1"><v>1</v></c></row>
  <row r="4" hidden="1"/>
```

For a trailing, otherwise empty hidden-row run, keep the final boundary. Hidden rows before later content retain their original representation.

Unsafe shortcut. Applying tail collapse inside the data region can change layout. This rule does not promise to preserve exterior blank-grid geometry.

Abbreviated XML. Style IDs are local.

Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

49 Quoted metadata is not another coordinate

Source

```
<row r="3">
 <c s='7'
  data-note='r="Z99" s="0"'
  r='A3'/>
</row>
```

=

Retained by the guard

```
<row r="3">
 <c s='7'
  data-note='r="Z99" s="0"'
  r='A3'/>
</row>
```

Scan attributes with quote awareness. The actual coordinate is A3 and the style is 7, independent of attribute order or embedded text.

Unsafe shortcut. A substring search can misread the quoted Z99 or s=0 text and remove a visibly formatted cell or move a bound.

Abbreviated XML. Style IDs are local.

50 Wanderungen zero tails

Source

```
 <cols>
  <col min="12" max="16384" hidden="1"/>
 </cols>
 <row r="11">
  <!-- visible cells retained -->
-  <c r="L11"><v>0</v></c>
-  <!-- zero cells through XFD11 -->
-  <c r="XFD11"><v>0</v></c>
 </row>
```

Guarded result

```
 <cols>
  <col min="12" max="16384" hidden="1"/>
 </cols>
 <row r="11">
  <!-- visible cells retained -->
 </row>
```

Tabelle 3.4 has 21 dense zero tails, L:XFD, in rows 11-30 and 32. Each has 16,373 cells. Admit only after hidden, layout and reference guards.

Unsafe shortcut. A hidden zero is still data. Keep mixed tables, the short 237-zero tail and the documented nonzero exceptions. A deliberate full-width constant matrix can meet this heuristic.

Abbreviated XML. Style IDs are local.

One representative, one native layout effect

Reduced fixture | installed Excel 16.0 build 20326 | full rebuild before each capture

One font witness preserves row layout

A2 has Arial 10. Empty B2 and C2 have Arial 18.

Source

```
<row r="2">
 <c r="A2">...</c>
 <c r="B2" s="1"/>
- <c r="C2" s="1"/>
</row>
```

Guarded result

```
<row r="2">
 <c r="A2">...</c>
 <c r="B2" s="1"/>
</row>
```

Original control **23.25 pt**

Font witness control			
10 pt text			
Next visible row			

One witness retained **23.25 pt**

Font witness control			
10 pt text			
Next visible row			

Both witnesses removed **12.75 pt**

Font witness control			
10 pt text			
Next visible row			

Native Excel vector export after full rebuild. Removing the repeated witness is safe here. Removing both moves the next visible row.

What was changed

The three input packages differ only in sheet1.xml. The source has two empty Arial 18 cells. The one-witness package removes C2. The negative control removes B2 as well. All use the same fonts, text and columns.

What Excel rendered

The source and one-witness vector exports have identical decoded pixels at 600 dpi. Both have row 2 height 23.25 pt. The negative control has row 2 height 12.75 pt and moves the next visible row upward. Its exported range is shorter.

Scope

These are constructed native controls, not the Kulturfinanzbericht screenshots. They illustrate the font-witness mechanism independently of the documented corpus case. They are not a full validation run or a measurement of shrink throughput.

A smaller file can expose a different fill

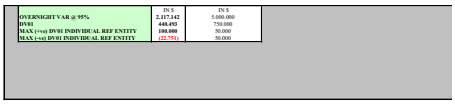
Retained Enron case ccde4294bab9094d7d73 | PL-Consolidated!A25:H37

An empty gray cell masks an inherited pink fill

Column B uses XF 51 (pink). Cell B31 uses XF 98 (gray).

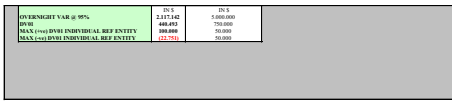
Original

```
<c r="B31" s="98"/>
```



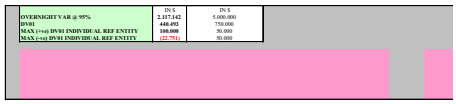
SheetShrink Default

```
<c r="B31" s="98"/>
```



Excel Check Performance

```
<!-- B31 removed -->
```



PL-Consolidated!A25:H37, vector export after full rebuild. Original and Default match pixel-for-pixel here. Removing the empty gray cells exposes the inherited pink background.

Original, rebuilt

	IN $	IN $
OVERNIGHT VAR @ 95%	**2.117.142**	5.000.000
DV01	**440.493**	750.000
MAX (+ve) DV01 INDIVIDUAL REF ENTITY	**100.000**	50.000
MAX (-ve) DV01 INDIVIDUAL REF ENTITY	**(22.751)**	50.000

Default, rebuilt

	IN $	IN $
OVERNIGHT VAR @ 95%	**2.117.142**	5.000.000
DV01	**440.493**	750.000
MAX (+ve) DV01 INDIVIDUAL REF ENTITY	**100.000**	50.000
MAX (-ve) DV01 INDIVIDUAL REF ENTITY	**(22.751)**	50.000

Check Performance, rebuilt

	IN $	IN $
OVERNIGHT VAR @ 95%	**2.117.142**	5.000.000
DV01	**440.493**	750.000
MAX (+ve) DV01 INDIVIDUAL REF ENTITY	**100.000**	50.000
MAX (-ve) DV01 INDIVIDUAL REF ENTITY	**(22.751)**	50.000

What the rejected shortcuts would change

Schematic illustrations | these are not Excel screenshots or measured coordinates

11 Contiguous centering span

Keep



Unsafe shortcut



One surviving blank cannot replace the complete span.

04 Local no-fill mask

Keep



Unsafe shortcut



Removing the K-cell mask exposes its blue parent fill.

25 Empty column boundaries

Keep



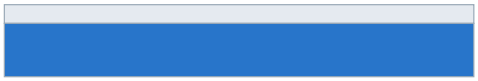
Unsafe shortcut



Changing widths moves borders even without values.

13 Visible exterior tail

Keep



Unsafe shortcut



A blue tail over white columns remains visible.

Rule-family coverage

Coverage inventory. Disabled and no-op paths are explicitly distinguished.

Payload and parser boundaries

Examples 01, 27, 29, 49

S1-S2, S4. Values, formulas, attributes, unchanged coordinates and explicit exclusions.

Inherited formatting and font layout

Examples 02, 03, 04, 05, 06, 07, 11, 12, 26, 30, 31, 38, 47

S4-S5. Font witnesses, local masks, fixed-height baselines, merges and centering spans.

Row representation proofs

Examples 08, 09, 10, 32, 33, 34, 46

S5. Dense and sparse promotion, gap restoration, dominant-style selection and fallback.

Tail rows and layout anchors

Examples 13, 14, 15, 16, 17, 18, 48

S5-S6. Content-free tails, visible masks, conditional scopes, drawing anchors and hidden boundaries.

Column coalescing

Examples 19, 25, 36, 37

S6. Compatible tails, font sentinels, visible border boundaries and uncovered gaps.

Admission and dependency-sensitive cleanup

Examples 20, 21, 22, 23, 35, 39, 40, 41, 42, 43, 44, 45, 50

S4, S7. Protection extent, conservative hidden admission, shared formulas, scoped object references and deferred re-cleaning.

Workbook metadata and package identity

Examples 24, 28

S1, S8-S9. Scoped aliases and calculation-chain maintenance. Existing style/string indices and unrelated members retain their identity.

All ten declared CleanupCause variants are accounted for. The examples cover the reviewed Default rule families plus explicitly labeled diagnostic phases. This is a review inventory, not a claim to enumerate every combination of formatting, formulas and Excel behavior.

Declared cleanup causes and their status

Coverage inventory. Disabled and no-op paths are explicitly distinguished.

PreprocessEmptySelfClosingCells

Examples No active Default transformation shown

Disabled by default. The optional preprocessor is not presented as a production Default reduction.

EmptyCellsWithoutContent

Examples 02, 03, 04, 05, 06, 07, 11, 12, 26, 29, 30, 31, 38, 47

Illustrated. Empty payloads are classified through their modeled formatting and layout effects.

EmptyHiddenCells

Examples 38, 39, 48

Illustrated. Hidden state alone does not imply removable content.

UnreferencedWhitespaceCells

Examples 23, 44

Explicit diagnostic scope. Default retains whitespace. Example 44 isolates the internal rewrite after candidate admission.

UnreferencedHiddenContentCells

Examples 20, 21, 39, 40, 41, 42, 43, 45, 50

Illustrated. Production admission is the conservative constant-tail rule. Example 45 is an injected-candidate phase test.

TrailingLayoutOnlyRows

Examples 13, 14, 15, 16, 17, 18, 48

Illustrated. The empty tail is distinguished from rows that position a later visual anchor.

EmptyRowAttributeCleanup

Examples No active Default transformation shown

Current helper is a no-op. remove_irrelevant_attributes_for_empty_rows returns the original attributes. No attribute-stripping transformation is claimed.

SheetDataRangeTailPruning

Examples 05, 14, 17, 18, 45

Illustrated. Range selection and later pruning must agree on the remaining anchors.

ExtraneousColumnRangeMerge

Examples 19, 25, 36, 37

Illustrated. Equal-fill and ordinary tail routes have separate eligibility and replacement predicates.

WorkbookDefinedNameCleanup

Examples 24

Known limitation retained. The current alias predicate matches name and scope, without proving equal formula bodies.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

01-content Content is not formatting

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | preserves_value_and_inline_string_elements_with_attributes

S4. Content tags are matched at their element-name boundary.

02-font-witness Keep one font witness

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | automatic_height_keeps_font_without_apply_font_or_row_style

Native control uses a reduced 18 pt fixture. The corpus regression is Kulturfinanzbericht, Impressum!B16, Arial MT 12 and no applyFont flag. Its documented heights are 15 and 12.75 pt.

03-merged-font A witness is local to a merge

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | merged_text_keeps_one_local_font_even_without_apply_font

Destatis Informationen zur Statistik!A6:H6. Native controls reported 43,767 changed pixels with all blanks removed, zero with B6 retained. Original images are retained on another machine.

04-no-fill No fill can be a visible override

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | unstyled_visual_masks_are_not_tentative_font_sentinels

PnP Sector Boilers, Fig 4 - SCR. An explicit XF with applyFill=1 and fillId=0 has the same guarded masking role.

05-redundant-no-fill Remove an override with nothing to clear

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | enron_august_no_fill_overrides_do_not_anchor_an_empty_sheet

Enron AUGUST, bundle 44ad86f1d9f249470fc6. The corpus case retained 21,947 unnecessary overrides before the fix.

06-row-xf-zero An implicit row style still overrides

excel-shrink-core/src/sheet/cells_regression_tests.rs

regression test | custom_format_without_style_uses_default_row_style

Raw Data F187:F190 is an independently restored Enron control. The AFFF manual Data rows 124-128 motivated the current row-format guard.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

07-column-font-mask Font witnesses cannot replace local masks

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | automatic_row_keeps_each_mask_of_an_unrepresented_hidden_column_font

Data.gov LCOE model, Installed Cost Overview. Documented row 14 changed from 14.25 to 15 pt before the correction.

08-row-promotion Move repeated formatting into the row

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | dense_full_width_row_compacts_identical_empty_styles_into_row_header

The full-width test removes 16,383 empty cells. No new style index is allocated.

09-promotion-gap Restore necessary exceptions at gaps

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | dense_row_preserves_internal_and_trailing_column_styles_on_repeat_cleanup

The regression uses gaps at E1, AD1, XFC1 and XFD1. It also checks a second cleanup pass.

10-automatic-promotion A cached height is not a fixed height

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | dense_row_promotion_requires_an_explicit_fixed_height

verkehrsunfaelle-zeitreihen, 3.8_(1). Native repair matched the 1,358 by 1,327 original capture. This drawing is a guard summary, not exact output XML. The excerpt shows the absence of row promotion. It does not enumerate ordinary cell cleanup.

11-center-span A centering span needs every blank

excel-shrink-core/src/sheet/output.rs

regression test | centered_text_keeps_only_its_contiguous_alignment_span

Data.gov Fig 10_New 10-29-19. The native counterfactual restored D2:G2, not a single sentinel.

12-protected-font Protection does not erase a font effect

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | exterior_protection_style_can_still_supply_a_required_font_sentinel

Enron 005d034f18d6dc340a2c had five one-pixel row-height changes. Active and inactive protection are covered by focused controls.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

13-white-tail Prune only a genuinely redundant fill

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | redundant_white_fill_and_font_override_do_not_anchor_an_empty_tail

Two Enron Feb cases ended at rows 486 and 485 after cleanup. Native white and blue controls matched selected original regions.

14-content-free A font alone need not anchor a tail

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | enron_nov_styleless_representatives_do_not_anchor_a_content_free_sheet

Enron NOV, bundles 44ad86f1d9f249470fc6 and a8bd950dff33454721b5. Exterior blank-grid geometry is outside this pruning contract.

15-border-mask A row header can clear column borders

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | borderless_tail_row_masks_a_visible_column_border_without_retaining_cells

Santavy Table Sup E motivated the initial-tail-range guard. No full row of cells is added.

16-border-height Empty row heights position visible borders

excel-shrink-core/src/sheet/output.rs

regression test | inherited_column_borders_keep_visible_tail_heights_only

The ordinary tail scan and content-boundary shortcut share this guard. Heightless and hidden redundant headers retain normal pruning.

17-conditional-layout Conditional ranges protect row layout

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | conditional_regions_preserve_empty_row_heights_without_restoring_cells

RN17B-RN30 Volatiles. Unknown conditional scope conservatively protects existing layout through the final Excel row.

18-drawing-anchor Objects extend the layout boundary

excel-shrink-core/src/sheet/output.rs

regression test | drawing_text_box_anchor_preserves_layout_row_after_last_cell

Missing or unsupported drawing information is a remaining review concern, not proof that no object exists.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

19-column-font Merge tail columns with font witnesses

excel-shrink-core/src/sheet/cols.rs

implementation anchor | tail_font_sentinels

Unknown styles, overlap, hidden columns, borders, uncovered gaps and crossing data boundaries restrict merging. Visible row borders also expose column widths.

20-hidden-tail Hidden constants need deferred approval

excel-shrink-core/src/sheet/hidden_content.rs

implementation anchor | MIN_TAIL_CELLS

4,096 is a conservative admission threshold, not proof of user intent. Deferred deletion re-cleans touched rows and invalidates stale tail anchors.

21-shared-formula Translate shared references by axis

excel-shrink-core/src/reference_scanner_sheet_scope_tests.rs

regression test | shared_formula_protects_shifted_hidden_inputs

Destatis verkehr-aktuell, 1.1.5. Unresolved groups, dynamic references or unknown scope disable reference-sensitive deletion.

22-chart-format A blank chart-source cell carries format

excel-shrink-core/src/reference_scanner/object_references.rs

regression test | chart_format_anchors_keep_only_the_first_source_cell

Current format-anchor discovery resolves direct worksheet references, not named or dynamic chart sources.

23-whitespace Whitespace can block text overflow

excel-shrink-core/src/shared_strings_tests.rs

regression test | detects_rich_text_shared_string_as_whitespace_only_when_all_runs_are_whitespace

Optional profiles have separate reference, protection and layout guards. ASCII whitespace classification does not cover every Unicode representation.

24-metadata Metadata has its own predicates

excel-shrink-core/src/workbook.rs

regression test | equivalent_alias_scope_is_independent_of_attribute_order_and_quotes

Calculation-chain cleanup separately removes entries for removed cells only when sheet mappings resolve. Unaffected styles, shared strings and binary assets retain their indices and content.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

25-border-width Column widths position repeated borders

excel-shrink-core/src/sheet/output.rs

regression test | visible_row_borders_preserve_empty_column_widths

This guard adds no cells and is propagated across parallel sheet chunks.

26-font-identity Compare font records, not just IDs

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | removes_repeated_empty_cells_when_column_font_records_are_identical

Style and font indices in the package are not renumbered. No general visual-equivalence test is inferred.

27-eligibility Unsupported input is an exclusion

excel-shrink-cli/src/pipeline_namespace_tests.rs

regression test | late_namespaces_abort_output_without_replacing_prior_results

Ordinary default namespaces and drawing extension prefixes are allowed. Strict SpreadsheetML, UTF-16 and explicit core prefixes have separate exclusion guards. The namespace URI is abbreviated in the drawing.

28-calculation-chain Prune stale chain entries by sheet

excel-shrink-cli/src/pipeline_regression_tests.rs

regression test | calc_chain_removes_missing_cells_for_processed_sheet_only

This is structural chain maintenance. Formula evaluation and replacement calculation caches remain Excel operations.

29-row-spans Trim spans without moving coordinates

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | keeps_one_empty_no_empty_cell_visible_style_footer_cell_after_content

Reduced from the row-16108 footer fixture. spans_use_only_current_row_cells_after_recleaning separately checks stale-state reuse.

30-empty-fixed-row Remove cells from an empty fixed row

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | removes_empty_no_empty_cell_visible_style_cells_when_row_already_has_custom_height

This fixture retains the original row attributes, including spans when present. It does not demonstrate a general attribute-stripping pass.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

31-fixed-baseline Fixed height still needs a font witness

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | fixed_height_text_row_keeps_explicit_default_font_with_no_fill

Pruefungen Hochschulen, TAB-04. This complements the automatic-height and merged-paragraph cases.

32-sparse-promotion Sparse promotion reuses a font-only XF

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | row_font_promotion_reuses_existing_font_xf_and_preserves_text

The fixture removes three empty cells and checks idempotency. Sparse promotion has different eligibility from the 1,024-occurrence dense routes.

33-uniform-fill Uniform fill promotion keeps merge cells

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | full_width_row_promotes_uniform_fill_and_preserves_merged_cells

The full-width regression removes 16,381 cells and retains C1:D1. Companion tests reject mismatched content fills and border/protection hazards.

34-dominance An ineligible majority still wins

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | row_font_promotion_does_not_substitute_a_promotable_minority_style

e18357b introduced explicit majority and tie counterexamples around the performance optimization. The panel shows the promotion decision, not the complete normal cleanup output.

35-protection-range Protection-only retention has a boundary

excel-shrink-core/src/sheet/output.rs

regression test | keep_in_data_range_keeps_inactive_protection_cell_inside_visible_data_range_only

The extent must account for later content. Separate tests cover active protection and KeepAll. Font and inherited-fill effects remain independent barriers.

36-column-coalescing Coalesce a compatible equal-fill tail

excel-shrink-core/src/sheet/cols.rs

regression test | collapses_visible_equal_fill_column_tail_with_no_borders

Widths are shortened for display. Font identities have their separate sentinel guard. The ordinary default-tail route has a distinct replacement predicate.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

37-column-gap Equal fill does not cover a gap

excel-shrink-core/src/sheet/cols.rs

regression test | equal_fill_does_not_paint_an_uncovered_gap

Other counterexamples retain tails when an inherited visible border, a hidden column or an unknown font style prevents the proof.

38-hidden-height Hidden values can affect visible height

excel-shrink-core/src/sheet/cells_regression_tests.rs

regression test | hidden_column_content_preserves_automatic_row_height

The regression enables both optional hidden-content and whitespace switches and checks the local admission barrier before deferred filtering.

39-short-hidden Small hidden tables are retained

excel-shrink-cli/src/pipeline_hidden_tail_tests.rs

regression test | hidden_tail_keeps_short_constant_suffix

The same module covers mixed hidden tables, nonzero details, full-width negative constants, gaps, formulas, text and value changes.

40-absolute-axes Shared formulas preserve absolute axes

excel-shrink-core/src/reference_scanner_sheet_scope_tests.rs

regression test | shared_formula_preserves_absolute_axes_and_other_sheet_scope

The scanner stores enclosing rectangles per reference. XML-escaped and apostrophe-containing sheet names have separate regression coverage.

41-sheet-scope Local references stay on their sheet

excel-shrink-core/src/reference_scanner_sheet_scope_tests.rs

regression test | local_reference_on_other_sheet_does_not_keep_candidate

Cross-sheet and local-name tests cover the positive direction. Selections, dimensions and print metadata are not automatically value consumers.

42-object-sources Charts and pivots have scoped consumers

excel-shrink-core/src/reference_scanner/object_references.rs

regression test | pivot_results_are_scoped_to_the_owning_sheet

Companion tests isolate pivot source ranges, chart series, linked drawing text/control formulas and whole-column references without per-cell expansion.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

43-dynamic-guard Unresolved sources block candidate deletion

excel-shrink-core/src/reference_scanner/object_references.rs

regression test | dynamic_object_references_disable_reference_based_removal

unresolved_named_pivot_sources_keep_candidates exercises the second source. Cell formulas also detect unresolved structured references and dynamic functions.

44-value-and-format Delete a value, keep its visible format

excel-shrink-cli/src/pipeline_regression_tests.rs

regression test | phase3_row_cleanup_preserves_stripped_whitespace_cell_with_visible_fill

The internal regression injects the candidate. It does not establish the broader future whitespace proof described in the specification.

45-reclean-tail Reconsider the tail after deferred deletion

excel-shrink-cli/src/pipeline_regression_tests.rs

regression test | phase3_trims_trailing_row_after_deferred_candidate_cleanup

This phase-level test injects an approved candidate at S41. Ordinary Default hidden-table admission remains the stricter constant-tail rule.

46-unknown-metadata Unfamiliar cell metadata blocks promotion

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | row_font_promotion_leaves_nonstandard_cell_metadata_on_the_normal_path

A separate test covers malformed coordinates, missing coordinates, malformed attributes and unknown styles. Those cases retain the local fallback rather than claiming normalization.

47-local-font-gap A visible font cannot mask another column

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | automatic_row_preserves_masks_without_retaining_redundant_font_bloat

With no column definitions, original unmasked gaps already supply the default font. The companion branch then allows redundant blank-font cleanup.

48-hidden-row-tail Collapse only the exterior hidden-row tail

excel-shrink-core/src/sheet/output.rs

regression test | trailing_hidden_tail_keeps_only_last_hidden_row

non_trailing_hidden_rows_are_not_collapsed_by_tail_cleanup is the contrasting fixture. Empty-row attribute cleanup currently returns the original attributes unchanged.

Source index and evidence limits

Pinned implementation 1b9ee98e3409 | full paths, anchors and hashes are in cases.json

49-quoted-attributes Quoted metadata is not another coordinate

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | cell_attributes_are_scanned_once_in_any_order_and_quoted_values_are_skipped

This is a parser-preservation guard supporting byte-oriented execution. It does not expand the declared XML eligibility to every valid serialization.

50-wanderungen Wanderungen zero tails

excel-shrink-cli/src/pipeline_hidden_tail_tests.rs

regression test | hidden_tail_still_removes_wanderungen_full_width_zero_tail

Source inspection confirms 343,833 zero records in these 21 tails. The historical cleanup investigation reports 376,573 removed zeros over two sheets and 243 retained nonzero values. These are historical output observations, not a fresh full-workbook visual certificate.